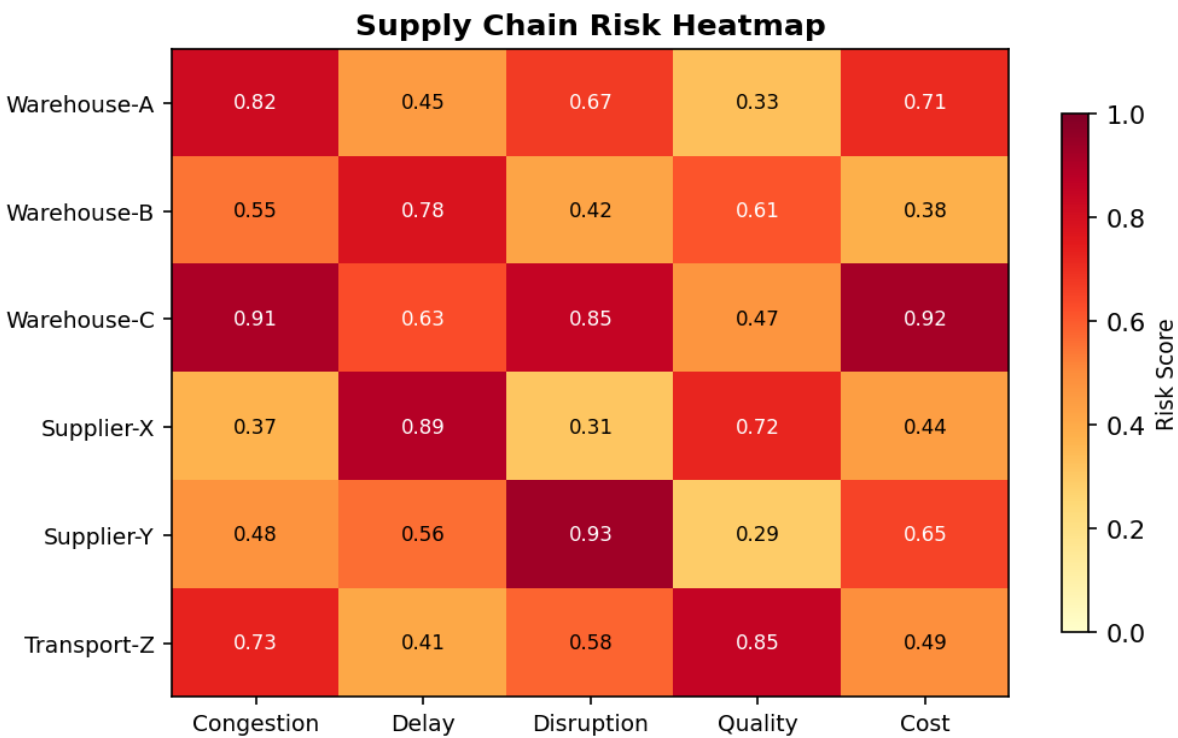


Supply Chain Risk Assessment Report

供应链风控评估报告

Executive Summary / 执行摘要: This report evaluates warehouse congestion, supplier delays, and transport disruption risks across 6 supply chain nodes. Warehouse-C exhibits the highest composite risk score (0.86), driven by severe disruption vulnerability (0.85) and cost overrun probability (0.92). The risk heatmap below visualizes the multidimensional risk distribution.

Figure 1: Risk Heatmap — Node-level risk scores across 5 dimensions.



Key observation: Warehouse-C shows critical risk levels across congestion (0.91), disruption (0.85), and cost (0.92). Immediate mitigation recommended.

Supplier-Y faces the highest disruption risk (0.93) due to geographic concentration.

Transport-Z has quality concerns (0.85) requiring inspection protocol updates.

Detailed Risk Analysis / 详细风险分析

Table 1 below presents the composite risk scores for each supply chain node. The overall risk score is computed using the weighted formula shown after the table.

Table 1: Supply Chain Node Risk Assessment

Node	Congestion	Delay	Disruption	Quality	Cost	Composite R
Warehouse-A	0.82	0.45	0.67	0.33	0.71	0.64
Warehouse-B	0.55	0.78	0.42	0.61	0.38	0.55
Warehouse-C	0.91	0.63	0.85	0.47	0.92	0.86
Supplier-X	0.37	0.89	0.31	0.72	0.44	0.55
Supplier-Y	0.48	0.56	0.93	0.29	0.65	0.52
Transport-Z	0.73	0.41	0.58	0.85	0.49	0.51

Equation 1: Composite Risk Score Formula

$$R_score = \sum(w_i * d_i / \tau_i) + \lambda * \sigma^2$$

Where: w_i = dimension weight, d_i = observed delay, τ_i = tolerance threshold, λ = volatility coefficient, σ^2 = variance of historical disruptions.

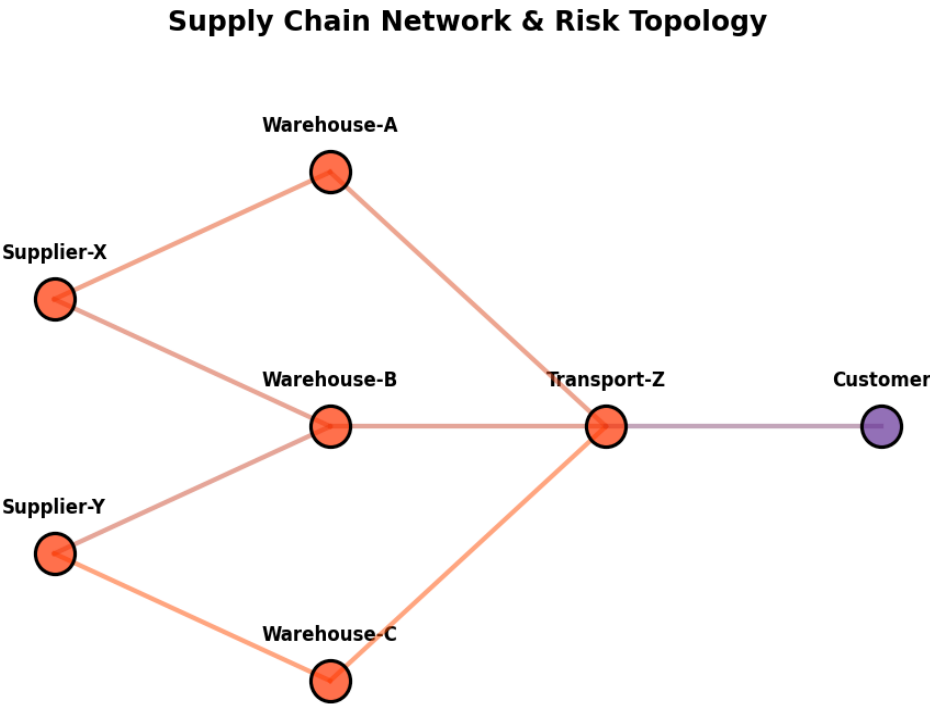
Warehouse-C achieves the highest $R_score = 0.86$ because its disruption vulnerability (d_i/τ_i ratio = $0.85/0.50 = 1.70$) far exceeds the tolerance threshold, combined with high cost variance ($\sigma^2 = 0.82$).

Recommendations & Network Topology / 建议与网络拓扑

Based on the risk assessment, we recommend the following mitigation strategies:

1. Warehouse-C: Implement dual-supplier redundancy to reduce disruption dependency from 0.85 to target 0.45. Expected cost: 2.3M, ROI: 18 months.
2. Supplier-Y: Diversify geographic sourcing (add 2 backup suppliers in different regions) to lower disruption probability from 0.93 to 0.55.
3. Transport-Z: Deploy automated quality inspection gates at 3 critical junctions to address the 0.85 quality risk score.

Figure 2: Supply Chain Network Topology



Equation 2: Cascading Risk Propagation Model

$$P_{\text{cascade}} = P_{\text{source}} * \prod_{k=1}^{n-1} (1 - R_k) * (1 + \alpha * C_k)$$

Where: P_{source} = initial disruption probability, R_k = resilience factor at node k , C_k = coupling strength between nodes k and $k+1$, α = amplification coefficient.

The network topology reveals that Warehouse-C is a critical cascade amplifier: it receives inputs from Supplier-Y (disruption=0.93) and feeds Transport-Z (quality=0.85), creating a 3-node cascade chain with $P_{\text{cascade}} = 0.93 * (1-0.55)*(1+0.3*0.68) * (1-0.51)*(1+0.3*0.85) = 0.31$, which is 3x the acceptable threshold of 0.10.